



Movie Genre Classification Project

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Dataset Selection

Initial options

- First explored a war/conflict dataset that had details like GDP
- Found inconsistencies / unreliable sources / likely fake data
- Needed a trustworthy dataset that could be used to predict verifiable conclusions

Why this dataset?

- Large dataset that had thousands of films that spanned decades
- Sizeable movie overviews that gave us enough text for NLP
- Multi-label genre tags enabled a classification prediction
- Flexible enough for Logistic Regression, XGBoost, LightGBM

Project Goal

- **Objective:** Explore factors that influence a movie's genre and ultimately use a movie's overview data to predict its genre
- **Method:** Clean data using OHE and embeddings, balance for underrepresented genres and test different models to analyze which one is the most accurate in predicting specific genres

Raw Data

Column	Type	Role
overview	Text	Primary NLP feature - needs to be extracted into vector representations
genres	Multi-label	Target Variable - needs to be cleaned and OHE
budget	Numeric	Heavily skewed right - needs to be transformed
release_date	Numeric	Extracted to be used

One Hot Encoding

```
mlb = MultiLabelBinarizer()

genre_ohe = pd.DataFrame(
    mlb.fit_transform(movies["genres_list"]),
    columns=mlb.classes_,
    index=movies.index
)

movies = pd.concat([movies, genre_ohe], axis=1)

movies.columns

Index(['adult', 'budget', 'id', 'imdb_id', 'original_language',
      'original_title', 'overview', 'popularity', 'release_date', 'revenue',
      'runtime', 'tagline', 'vote_average', 'vote_count', 'genres_list',
      'languages_list', 'production_companies_list',
      'production_countries_list', 'Action', 'Adventure', 'Animation',
      'Comedy', 'Crime', 'Documentary', 'Drama', 'Family', 'Fantasy',
      'Foreign', 'History', 'Horror', 'Music', 'Mystery', 'Romance',
      'Science Fiction', 'TV Movie', 'Thriller', 'War', 'Western'],
      dtype='object')
```

- Converted genre lists into multi-label format
- Used MultiLabelBinarizer to create binary features
- One column per genre (multi-hot encoding)
- Each movie represented as a 0/1 genre vector
- Used as targets for classification models

Overview Embeddings

- Used each movie's **overview** as the primary NLP feature
- Converted overview text into dense semantic vectors using transformer-based embedding models
- Tested all-mpnet-base-v2, all-MiniLM-L6-v2, and intfloat/e5-large-v2
- MiniLM-L6-v2 was faster, but it captured less of the deeper semantic context needed for our task
- Among the models we tested, e5-large-v2 produced the most useful representation for genre prediction because movie overviews behave like short passages, where storyline, tone, and theme matter more than isolated keywords

```
model = SentenceTransformer('intfloat/e5-large-v2', device=device)

texts = ["passage: " + t for t in movies['overview'].tolist()]

embeddings = model.encode(
    texts,
    show_progress_bar=True,
    device=device
)

embeddings = np.asarray(embeddings)
print('Embeddings shape:', embeddings.shape)
```

Text → Embedding
Vector → Genre
Classifier

Feature Engineering

- Created additional numeric features from metadata to complement text embeddings
- Applied log transforms to budget and revenue because both variables were heavily right-skewed
- Converted release_date into a datetime field for later temporal EDA
- Combined semantic text features with structured metadata in downstream experiments

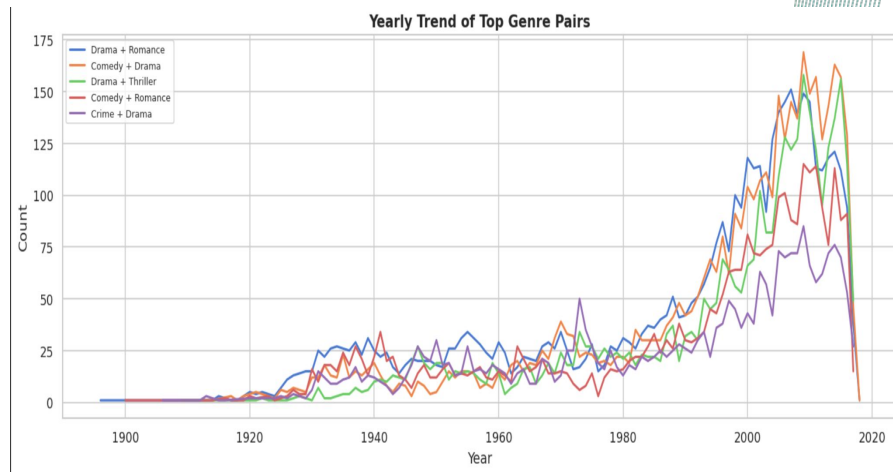
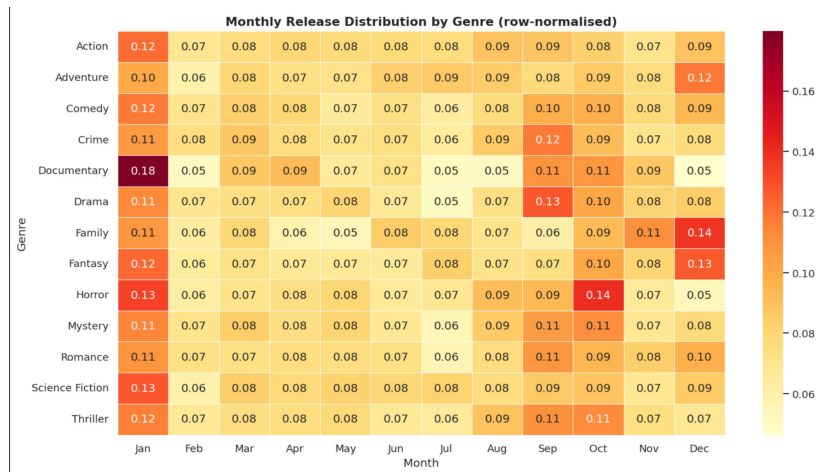
```
movies['log_revenue'] = np.log(movies['revenue'].fillna(0).astype(int) + 1)  
movies['log_budget'] = np.log(movies['budget'].fillna(0).astype(int) + 1)
```

Why log transform?

- Reduces the effect of extreme outliers and makes skewed numeric variables more stable for modeling

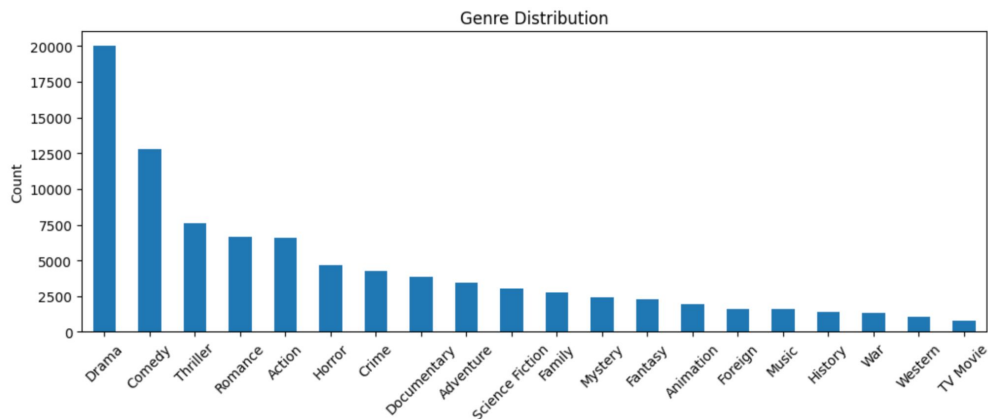
EDA: Release Date x Genre Trends

- Extracted **year, month, quarter, and season** from each movie's release date
- Analyzed how genre frequency changed over time using yearly and seasonal views
- Also explored monthly concentration and genre co-occurrence patterns
- Temporal EDA helped us understand how genre labels varied beyond simple counts



EDA: Genre Distribution

- The dataset was highly imbalanced across genres
- Drama and Comedy had the most samples, while TV Movie, Western, and War were much rarer
- This imbalance made the multi-label classification task harder, especially for rare genres
- This led us to apply a **genre-frequency** cutoff before final modeling; we discuss its performance impact later



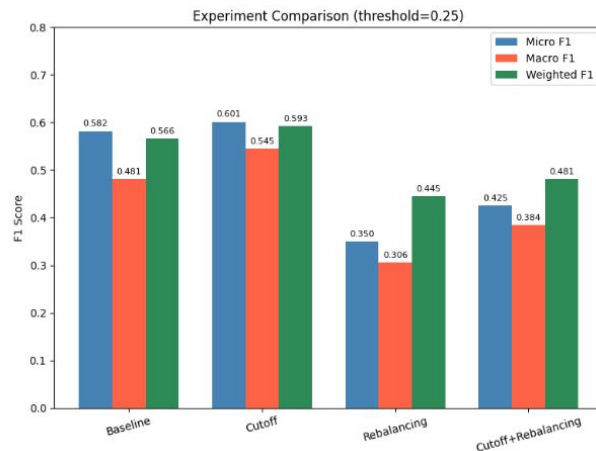
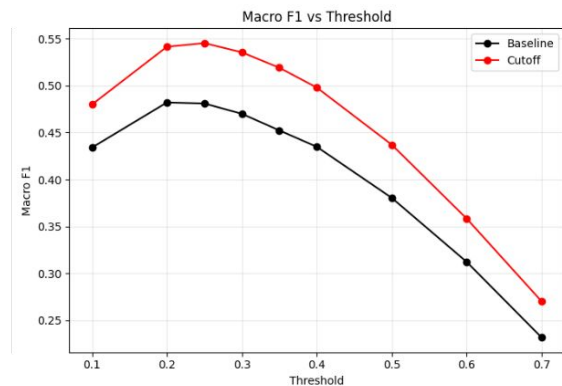
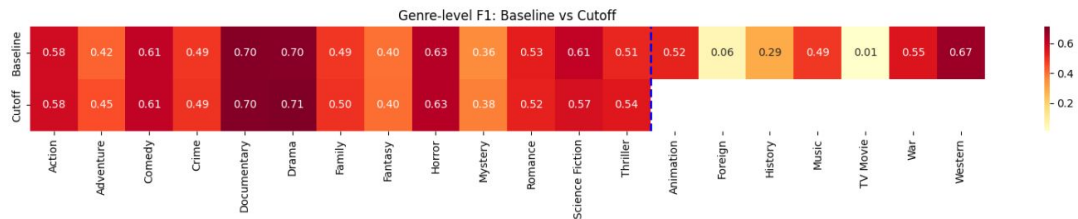
- Drama: **20,023**
- Comedy: **12,806**
- Thriller: **7,586**
- TV Movie: **751**
- Western: **1,035**
- War: **1,310**

Genre Imbalance Treatment

- Solution 1 : Cutting off genres under 2000
- Solution 2 : Weight rebalancing
- Solution 3: Cut off + Weight Rebalancing
- Reduced from 20 to 14 genres -> compromising
- Over-weighting rare genres confused

the model, hurting prediction on dominant genres

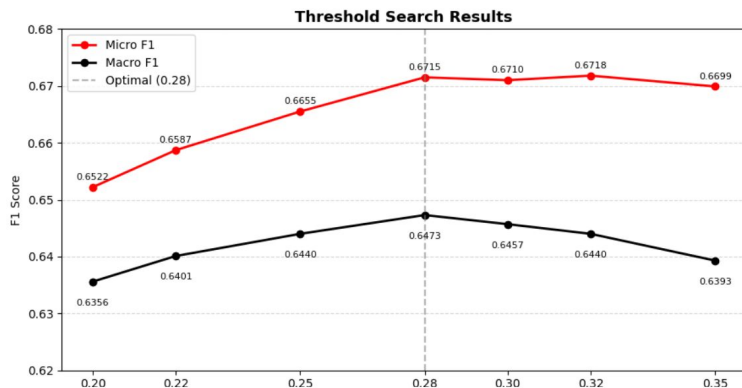
Result : Cutting off is the best



Feature comparison

All of the models were used in Logistic Regression

- $F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) \rightarrow \text{imbalance!}$
- Embedding : title+tagline+overview
- Not adding date is better
- Threshold : 0.28
- Cutoff threshold : 3000



Final Comparison (Macro F1)

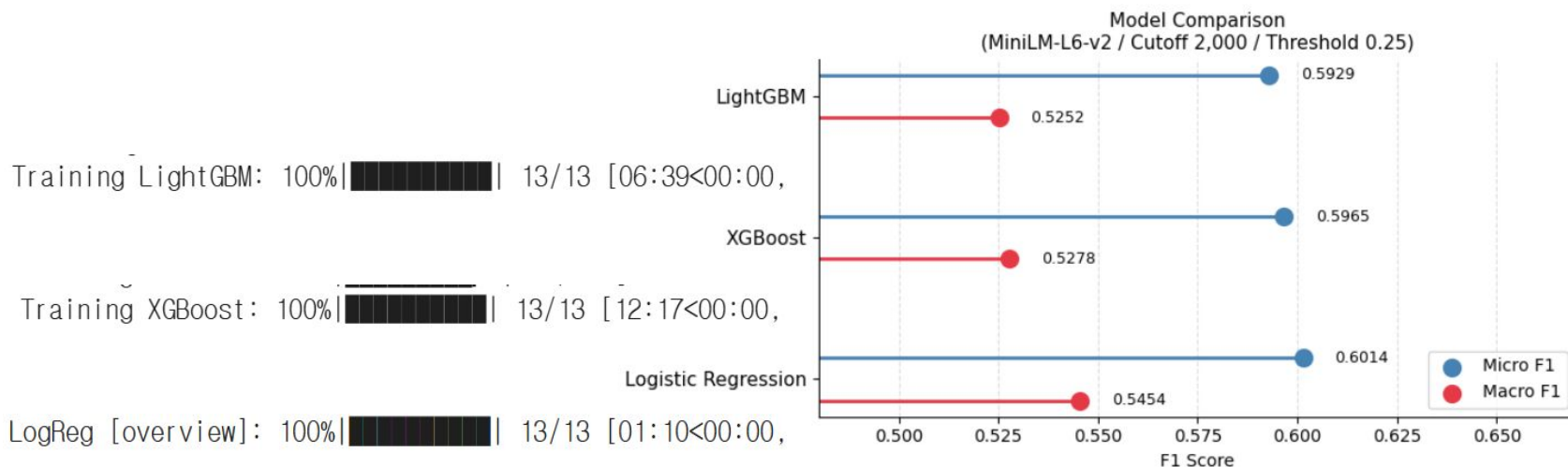
LogReg_overview	Macro: 0.5463 Micro: 0.6018
LogReg_title+overview	Macro: 0.5473 Micro: 0.6023
LogReg_tagline+overview	Macro: 0.5528 Micro: 0.6077
LogReg_title+tagline+overview	Macro: 0.5559 Micro: 0.6098

설정	Threshold	Micro	Macro
e5 + overview (baseline)	0.25	0.6655	0.6440
e5 + overview + date	0.28	0.6571	0.6290
e5 + overview + date	0.25	0.6534	0.6284

Modeling

All models tested on MiniLM-L6-v2 embeddings / Cutoff 2,000 / Threshold 0.25

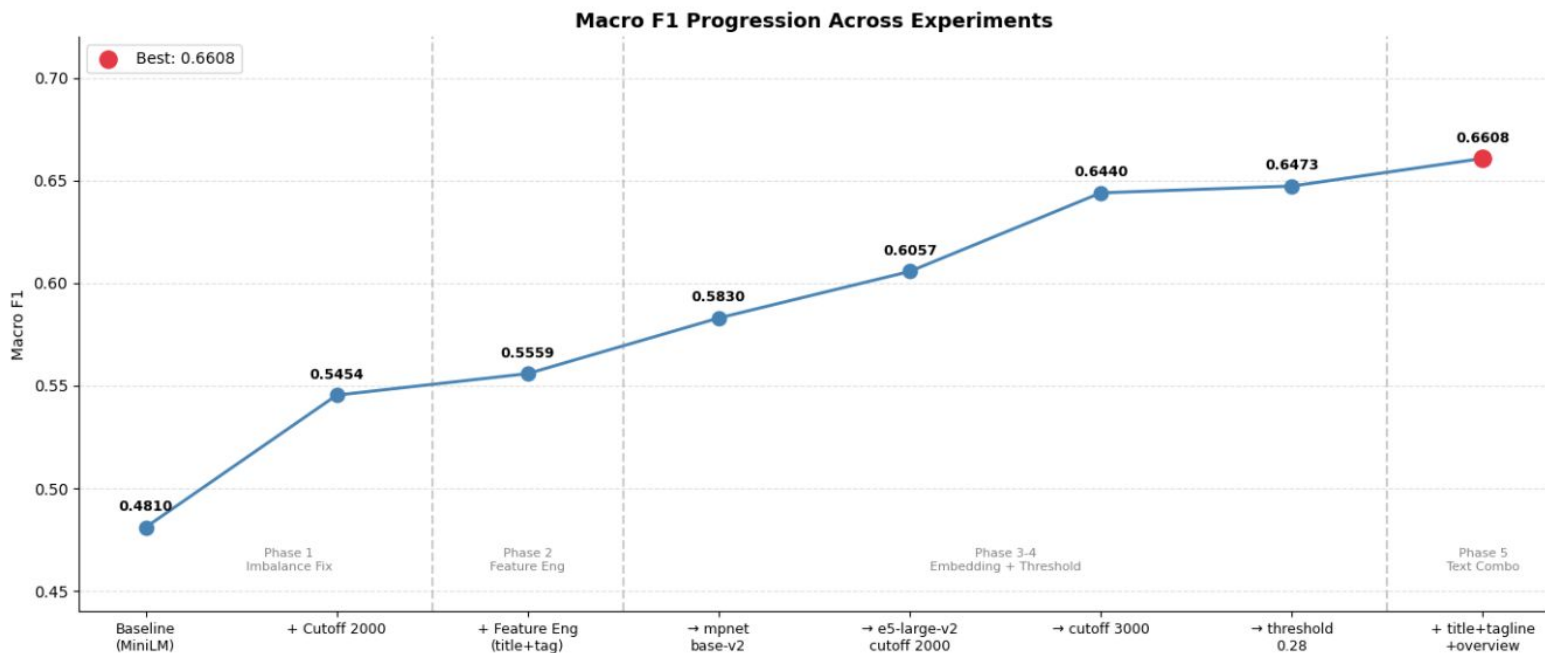
- **Logistic Regression** did the best performance with shortest time
- SentenceTransformer embeddings already separate genres linearly — complex models like XGBoost add no benefit



Result

**e5-large + cutoff 3000 + threshold + 0.28 + title + tagline + overview embedding
+ LogRegression**

= Micro: 0.6825 | Macro: 0.6608



Why Logistic Regression Outperformed Tree Models

High-Dimensional Embeddings

e5-large-v2 produces 1,024-dimensional vectors. Class boundaries are already approximately linear in this space — logistic regression is built for exactly this.

Speed Advantage

Logistic Regression trained in ~1 min. XGBoost took 12+ minutes and LightGBM 6+ minutes— with no accuracy gain. Complexity without benefit.

Why Tree Models Struggle

XGBoost & LightGBM split one feature at a time. With 1,024 dimensions they need exponentially more splits to approximate what logistic regression captures in one hyperplane.

Linear Genre Separability

Transformer embeddings map semantic meaning to geometry. Genre clusters in embedding space are linearly separable — matching logistic regression's inductive bias perfectly.

Model Comparison

MiniLM / Cutoff 2k / Threshold 0.25

Logistic Regression

Macro F1: 0.6608

Micro F1: 0.6825

LightGBM

Macro F1: 0.5252

Micro F1: 0.5929

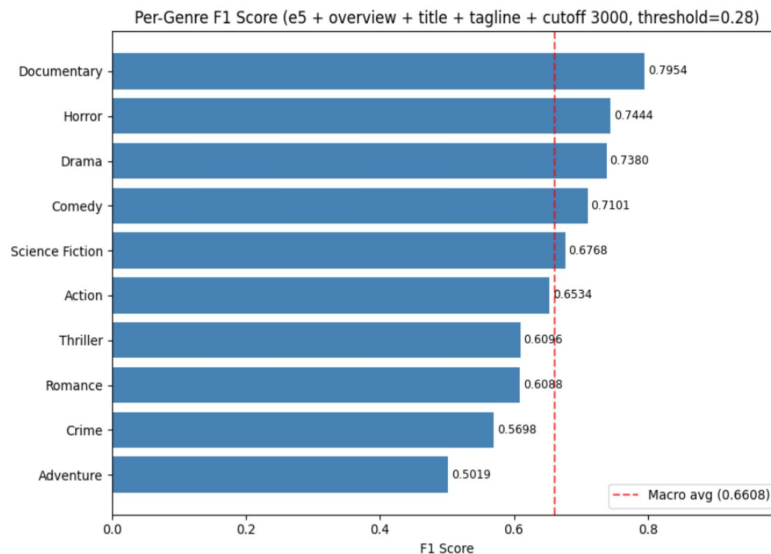
XGBoost

Macro F1: 0.5278

Micro F1: 0.5965

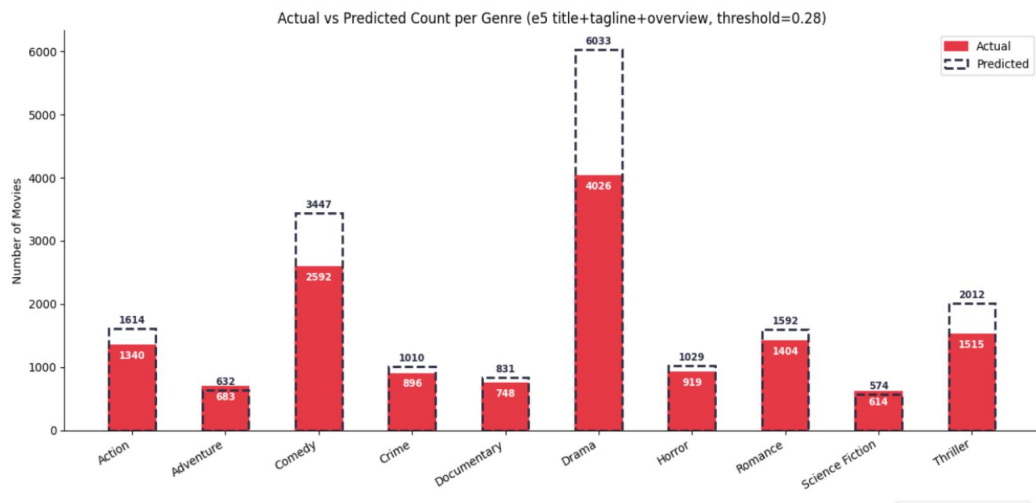
Results: Per-Genre Model Performance

Performance per genre



Best genres: Documentary (0.795), Horror (0.744), Drama (0.738) — distinctive language in overviews makes these easiest to separate.

Drama bias: Drama has 4,026 actual samples but model predicts only 6,033 — systematic over-prediction for the largest class, pulling the macro average down.



Hardest genre: Adventure (0.502) — highly overlapping with Action, Drama & Sci-Fi; the model under-predicts it most severely.

Model threshold: 0.28 threshold chosen to balance precision/recall tradeoff; lower values over-predict, higher values miss rare genres.

Next Steps & Future Work

More Data

1

Drama & Comedy still dominate even after the 3,000-cut-off — more balanced corpora (e.g., streaming catalogs) could reduce this skew

2

Scrape additional metadata: director, cast, production studio — these carry strong genre signals not captured in text

3

Augment rare-genre samples using paraphrase generation to boost Adventure, Crime, and Romance counts

Better Embeddings

1

Test newer models: GTE-large, BGE-M3, or OpenAI text-embedding-3-large for richer semantic representations

2

Fine-tune e5-large-v2 directly on movie overviews with contrastive genre labels — domain adaptation could push Macro F1 above 0.72

3

Experiment with ensemble of embeddings (title vs overview vs tagline as separate towers)

Modeling Approaches

1

Try multi-label neural classifiers (BERT + classification head) that jointly learn all genres — may capture genre co-occurrence better

2

Explore label-propagation or graph-based approaches to model genre correlations (Drama+Romance often co-occur)

3

Per-genre threshold tuning: Adventure may need a lower cutoff (~0.20) to recover recall on underrepresented labels

The background features abstract, flowing wavy lines in shades of teal and orange, creating a sense of movement and depth. These lines are composed of many thin, overlapping strokes that form a ribbon-like effect.

Thanks!